

Emma Dann

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Education

PhD in Biological Sciences (Computational Biology) – University of Cambridge, UK (Oct 2019 – Oct 2023)

Advisors: Prof. Sarah A. Teichmann, Dr. John C. Marioni

Thesis title: *Discovering variation from cell atlases: comparative methods for single-cell genomics*

MSc in Cancer, Stem Cells and Developmental Biology (Bioinformatics profile) – Utrecht University, NL (Sep 2017 – Aug 2019)

BSc in Biomolecular Sciences and Technology – University of Trento, IT (Sep 2014 – July 2017)

Academic appointments

Post-doctoral fellow – Stanford University, Palo Alto, US (May 2024 – present)

Visiting post-doctoral fellow – Gladstone Institute for Genomic Immunology, San Francisco, US

Hosted by Jonathan Pritchard and Alex Marson. Linking population genetic variation to immune traits with single-cell perturbation screens.

Graduate researcher – Wellcome Sanger Institute, Cambridge, UK (Oct 2019 – Jan 2024)

Supervised by Sarah Teichmann and John Marioni. Statistical methods for comparative analysis of single-cell omics data.

Research intern – European Molecular Biology Laboratory, Heidelberg, DE (Sep 2018 – Aug 2019)

Supervised by Wolfgang Huber. Analysis of multi-omic profiles of perturbations for functional characterization of drug treatments and target discovery.

Research intern – KNAW Hubrecht Institute, Utrecht, NL (Sep 2017 – Aug 2018)

Supervised by Alexander Van Oudenaarden. Biophysical modelling of whole-genome amplification via random priming in single-cell bisulfite sequencing experiments.

Research intern – Center for Integrative Biology, University of Trento, IT (Feb 2017 – July 2017)

Supervised by Francesca Demichelis. Epigenetic biomarker discovery from prostate cancer whole-methylome data.

Publications

► denotes manuscripts as leading author, * denotes equal contribution

Ongoing and Pre-print Manuscripts

► Zhu R.*, **Dann E.***, Yan J., Retana J.R., Goto R., ... , Pritchard J.K., Marson A. (2025) Genome-scale perturb-seq in primary human CD4+ T cells maps context-specific regulators of T cell programs and human immune traits. *bioRxiv* | [Full text](#)

Deconinck L, Zappia L, Cannoodt R, Morgan M, **scverse core**, ... , Saeys Y. (2025) *anndataR* improves interoperability between R and Python in single-cell transcriptomics. *bioRxiv* | [Full text](#)

Pett J.P., Prete M., Pham D., England N., ... , **Dann E.**, ... , Teichmann S.A., Meyer K. (2025) An atlas of TF driven gene programs across human cells. *bioRxiv* | [Full text](#)

Heumos L., Ji Y., ... , **Dann E.**, ..., Theis F.J. (2024) Pertpy: an end-to-end framework for perturbation analysis. *bioRxiv* | [Full text](#) | [python package](#)

► **Dann E.***, Teeple E.*, Elmentaite R., ... , Rinaldis E., Teichmann S.A. (2024) Estimating the impact of single-cell RNA sequencing of human tissues on drug target validation., *medRxiv* | [Full text](#) | [code](#)

Peer-reviewed Manuscripts

- Ota M., Spence J.P., Zeng T., **Dann E.**, Milind N., Marson A., Pritchard J.K. (2025) Causal modeling of gene effects from regulators to programs to traits: integration of genetic associations and Perturb-seq. *Nature* | [Full text](#)
- Luecken M., Gigante S., Burkhardt D., Cannoodt R., ... , **Dann E.**, ... , Saeys Y., Theis F.J., Krishnaswamy S. (2025) Defining and benchmarking open problems in single-cell analysis. *Nature Biotechnology* | [Full text](#)
- Schuster V., **Dann E.**, Krogh A., Teichmann S.A. (2024) multiDGD: A versatile deep generative model for multi-omics data. *Nature Communications* | [Full text](#) | [python package](#)
- Yayon N., Kedlian V., Boehme L., ... , **Dann E.**, ... , Uhlmann V., Notarangelo L.D., Germain R.N., Radtke A.J., Marioni J.C., Taghon T., Teichmann S.A. (2024) A spatial human thymus cell atlas mapped to a continuous tissue axis. *Nature* | [Full text](#)
- Oliver A.J., Ni H., ... , **Dann E.**, ... , Elmentaite R., Teichmann S.A. (2024) Single-cell integration reveals metaplasia in inflammatory gut diseases. *Nature* | [Full text](#)
- Missarova A., **Dann E.**, ... , J.C. Marioni (2023) Leveraging neighborhood representations of single-cell data to achieve sensitive DE testing with miloDE. *Genome Biology* | [Full text](#) | [R package](#)
- Sumanaweera D., Suo C., Muraro D., **Dann E.**, ... , Teichmann S.A. (2023) Gene-level alignment of single cell trajectories. *Nature Methods* | [Full text](#) | [python package](#)
- Barnes J. L., He P., ... , **Dann E.**, ... , Teichmann S.A., Meyer K. B., Nikolic M. Z. (2022) Early human lung immune cell development and its role in epithelial cell fate. *Science Immunology* | [Full text](#)
- **Dann E.**, Cujba A.M., Oliver A., Meyer K., Teichmann S.A., Marioni J.C. (2023) Precise identification of cell states altered in disease with healthy single-cell references, *Nature Genetics* | [Full text](#) | [code](#) | [python package](#)
- Botting R.A., Goh I., ... , **Dann E.**, ... , Teichmann S.A. , Haniffa M. (2023) Yolk sac cell atlas reveals multiorgan functions during human early development. *Science* | [Full text](#)
- Suo C., Polanski K., **Dann E.**, ... , Tuong Z.K., Clatworthy M., Teichmann S.A. (2023) Dandelion uses the single-cell adaptive immune receptor repertoire to explore lymphocyte developmental origins, *Nature Biotechnology* | [Full text](#) | [python package](#)
- Heumos L., ... **Single-cell Best Practices Consortium**, ... , Theis F.J. (2023) Best practices for single-cell analysis across modalities. *Nature Review Genetics* | [Full text](#)
- Suo C.*, **Dann E.***, ... , Haniffa M., Teichmann S.A. (2022) Mapping the developing human immune system across organs. *Science* | [Full text](#) | [code](#) | [data](#)
- Lalchand V.*, Ravuri A.*, **Dann E.***, ... , Teichmann S.A., Lawrence N.D. (2022) Modelling Technical and Biological Effects in scRNA-seq data with Scalable GPLVMs. *Proceedings of the 17th Machine Learning in Computational Biology meeting* | [Full text](#)
- He P., Lim K., Sun D., ... , **Dann E.**, ... , Meyer K.B., Rawlins E.L. (2022) A human fetal lung cell atlas uncovers proximal-distal gradients of differentiation and key regulators of epithelial fates, *Cell* | [Full text](#)
- Kleshchevnikov V., Shmatko A., **Dann E.**, ... , Stegle O., Bayraktar O.A. (2022) Cell2location maps fine-grained cell types in spatial transcriptomics. *Nature Biotechnology* | [Full text](#) | [python package](#)
- **Dann E.**, Henderson N.C., Teichmann S.A., Morgan M.D., Marioni J.C. (2022) Differential abundance testing on single-cell data using k-nearest neighbor graphs, *Nature Biotechnology* | [Full text](#) | [R package](#) | [python package](#)
- Jardine L., Webb S., ... , **Dann E.**, ... , Göttgens B., Roberts I., Teichmann S. and Haniffa M. (2021) Blood and immune development in human fetal bone marrow and Down syndrome, *Nature* | [Full text](#)
- Elmentaite R., Kumasaka N., Roberts K., Fleming A., **Dann E.**, ... , James K.R., Teichmann S.A. (2021), Cells of the human intestinal tract mapped across space and time, *Nature* | [Full text](#)
- Stephenson E., Reynolds G., Botting R.A., Calero-Nieto F.J., Morgan M.D., Tuong Z.K., Bach K., Sungnak W., ... , **Dann E.**, ... , Duncan C.J.A, Smith K., Teichmann S.A., Clatworthy M.R., Marioni J.C., Gottgens B., Haniffa M. (2020) Single-cell multi-omics analysis of the immune response in COVID-19, *Nature Medicine* | [Full text](#)

Beltran H., Romanel A., ... , **Dann E.**, ... , Benelli M., and Demichelis F. (2020) Circulating Tumor DNA Profile Recognizes Transformation to Castration-Resistant Neuroendocrine Prostate Cancer, *Journal of Clinical Investigation* | [Full text](#)

Fellowships & awards

Tahoe Deep Dive Hackathon, first prize winner (May 2025)

Awarded \$20,000 for best project on prediction of clinical outcomes from large-scale drug perturbation datasets.

Helen Hay Whitney fellowship (Nov 2024 – present)

Postdoctoral fellowship, Awarded \$229,500 over 3 years

European Molecular Biology Organization (EMBO) long-term fellowship (Jul 2024 - present)

Postdoctoral fellowship, non-stipendiary

Wellcome Sanger Institute PhD studentship, Wellcome Trust, UK (Oct 2019 - Oct 2023)

PhD fellowship, Awarded ~ £100,000 plus tuition over 4 years

U/Select honours programme Utrecht University, NL (Jan 2018 - Oct 2019)

Master's honors program, including €2000 travel grant for international research internship

Merit Award for excellent graduates University of Trento, IT (Oct 2017)

Prize for excellent graduates, awarded €3000

Grants

Contributed to NIH R01 grant (2R01HG008140) (2025)

Title: Integration of genetic association mapping and functional data to elucidate genetic mechanisms of disease

PIs: Jonathan K. Pritchard, Alex Marson

Reviewed as 1st percentile score

Contributed to ERC Advanced Grant (ThyDESIGN) (2022)

Title: Learning from the thymic human cell atlas for T cell engineering

PIs: Sarah A. Teichmann

Awarded €2.5M over 5 years

Presentations

Conference oral presentations

Apr 2025 - Chan Zuckerberg Initiative Cell Science Meeting, Chicago (US)

Nov 2022 - Machine Learning in Computational Biology conference, virtual

Jun 2022 - Single Cell Biology conference, Wellcome Genome Campus, Hinxton (UK)

May 2022 - The Biology of Genomes, Cold Spring Harbor Laboratory (US)

Oct 2021 - Chan Zuckerberg Initiative Single Cell Biology Meeting, virtual

Sep 2021 - Next Generation Genomics Meeting, virtual

Mar 2021 - CZI Seed Networks Computational biology meeting, virtual

Invited research seminars

Oct 2025 - Ultima Genomics user group session @ American Society for Human Genetics, Boston (US)

Sep 2025 - scverse community meeting, virtual

Jul 2025 - AI + biomedicine seminar, Stanford (US)

Sep 2024 - BRAID Genentech, South San Francisco (US)

Sep 2024 - Human Genetics department, University of Chicago, Chicago (US)

Sep 2024 - SQLIFTS Lecture Series, Northwestern University, Chicago (US)

Mar 2024 - Human Technopole, Milan (IT)

Dec 2023 - Google DeepMind, London (UK)

Jul 2023 - Sanofi Precision Medicine & Computational biology, Cambridge (US)

Jun 2023 - Chan Zuckerberg Initiative, Redwood City (US)

Apr 2023 - Cambridge AI club for biomedicine, Milner Institute, Cambridge (UK)

Dec 2022 - UMC Utrecht single-cell genomics meeting, virtual

Sep 2022 - UK Conference of Bioinformatics and Computational Biology, virtual
Jul 2022 - Computational Health Center seminar, Helmholtz Munich (DE)
Apr 2022 - Maxwell Society Annual Conference, King's College London (UK)
Apr 2022 - NYU Langone single cell journal club, virtual
Mar 2021 - Cambridge Center for Physical Biology Single Cell Symposium, virtual

Guest lectures

Aug 2024 - Why there's never been a better time to study human biology, [Girls in Research](#) program (virtual)
Aug 2023 - Intro to single-cell genomics series, University of Sri Jayewardenepura (virtual)
Mar 2023 - Introduction to multiomics data integration and visualisation, EMBL-EBI training course (UK)
Apr 2022 - Best practices and open challenges in single-cell genomics data analysis, Maxwell Society Annual Conference, King's College London (UK)

Conference poster presentations

Oct 2025 - American Society for Human Genetics conference, Boston (US)
Jul 2023 - Human Cell Atlas General Meeting, Toronto (CA)
Oct 2022 - Single Cell Genomics conference, Utrecht (NL)
Jun 2021 - Human Cell Atlas General Meeting, virtual

Teaching & mentoring

Computational research assistant recruitment and mentoring, Marson lab (Fall 2025 - present)

PhD rotation student recruitment and mentoring, Pritchard lab (Winter 2025 - present)

Stanford & Tanzania Human Genetics Organization virtual training program (Spring 2025)
One-on-one virtual mentoring program on research project for students from Tanzania

PhD rotation student recruitment and mentoring, Teichmann lab (Fall 2021 - Fall 2023)

Systems biology: From large datasets to biological insight, EMBL-EBI training course (Jul 2022)
Invited instructor for single-cell multi-omics module ([course material](#))

Advanced topics in Single Cell Genomics, Swiss Institute for Bioinformatics course (Apr 2022)
Invited instructor for multi-omics module ([course material](#))

Advanced topics in Single Cell Genomics, SciLifeLab / Swiss Institute for Bioinformatics Summer School (Aug 2021)
Invited instructor for multi-omics module ([course material](#))

Leadership & outreach

Core team member, scverse consortium (Feb 2024 – present)

I am part of the governance team of the consortium, overseeing the development and maintenance of foundational software tools for omics data analysis. I am especially involved in community engagement activities and partnerships

- Lead organizer of the [scverse annual conference](#) and [community hackathons](#)
- Contributor to open-source [tutorials](#) and [resources](#) for newcomers to single-cell genomics analysis in python

Competition judge, Open Problems Single-cell perturbations Kaggle competition (Fall 2023)

Served as judge for NeurIPS-affiliated machine learning challenge focused on predicting drug perturbation responses.

Co-organizer, Strengthening integration pipelines session, Human Cell Atlas general meeting (July 2023)
Co-organized and led panel discussion on cell atlas construction and usage.

Member of the selection panel, Sanger prize competition (Mar 2023)

The Sanger Prize competition funds research internships at Sanger for undergraduate students from LMI countries. I was a member of the Grad Panel, evaluating essays from applicants.

Co-organizer Open Problems in Single-Cell Analysis Jamboree (April 2021)
Led task for benchmarking of compositional analysis methods for scRNA-seq.

Co-organizer, EBI-Sanger-Cambridge PhD Symposium (Feb 2020)
I was in charge of the scientific program and sponsorship for the PhD program symposium.

Board member, Open Wet Lab (OWL) (Oct 2015 – Jan 2018)
OWL is the first biohacking organization in Italy. I was in charge of the association's budget and directed the publication of OWL's blog and scientific outreach articles for local newspapers.

Peer review

Journals: Nature (1 paper), Cell (1* paper), Nature Biotechnology (2* papers), Nature Genetics (1 paper), Bioinformatics Advances (1 paper), OUP Bioinformatics (1 paper)
*Assisting a senior reviewer

Conferences: Machine Learning in Computational Biology

Grants/fellowships: Chan Zuckerberg Initiative Single Cell Biology Data Insights, Christian Doppler Research Association Transfer.S2S grants

Software

pertpy |  [scverse/pertpy](#) | contributor | perturbation analysis toolkit for the scverse ecosystem.

miloR |  [MarioniLab/miloR](#) | primary developer | R/Bioconductor package for the identification of differential cell abundance in KNN graphs for single-cell data

milopy |  [emdann/milopy](#) | primary developer | python package for the identification of differential cell abundance in KNN graphs for single-cell data

genomic-features |  [scverse/genomic-features](#) | primary developer | Genomic annotations using Bioconductor resources in Python.

oor_benchmark |  [MarioniLab/oor_benchmark](#) | primary developer | python framework to benchmark workflows for detection of out-of-reference cells in comparative single-cell analysis

cell2location |  [BayraktarLab/cell2location](#) | contributor | python package for mapping of tissue cell types via integration of single cell RNA-seq and spatial transcriptomics data

Industry experience

Data science consultant – Ensocell Ltd (Oct 2023 – Mar 2024)
Target discovery platform and analysis of single-cell data

References

Jonathan K. Pritchard, PhD – Post-doctoral advisor
Bing Professor of Population Studies, Departments of Genetics and Biology, Stanford University
pritch@stanford.edu

Alexander Marson, PhD – Post-doctoral advisor
Director, Gladstone-UCSF Institute of Genomic Immunology
Professor, Department of Medicine, UCSF
alexander.marson@ucsf.edu

Prof Sarah A. Teichmann, PhD – PhD advisor
Professor, Cambridge Stem Cell Institute & Department of Medicine, University of Cambridge (UK)
sat1003@cam.ac.uk

John C. Marioni, PhD – PhD advisor

Senior Vice President and Head of Computation, Genentech Research and Early Development
marioni.john@gene.com

Muzlifah Haniffa, PhD – Graduate collaborator

Head of Cellular Genetics, Wellcome Sanger Institute, UK
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